CAR SALES DATA PIPELINE – PROCESS FLOW

1 DATA SOURCE

car_sales.csv

- Source: Local file
- Format: CSV
- Raw car sales data

OUTPUT
Raw CSV File

2 LOAD INTO DATABRICKS

- Create database (if not exists)
- Upload CSV to DBFS
- Create table from CSV

OUTPUT
Raw Table
workspace.default.car_sales

3 DATA TRANSFORMATION & CLEANING

- Deduplication (is_duplicate flag)
- Standardize text fields (make, model, etc.)
- Parse date (saledate_clean)
- Impute missing values
  - condition_imputed
  - odometer_fixed
- Round numeric values
- Create data quality flags
- Create validation flag

OUTPUT
Cleaned & Enriched Table
workspace.default.car_sales_clean

4 DATA VALIDATION & QUALITY CHECKS

- Check total record count
- Check duplicates
- Check missing values
- Review data quality flags
- Ensure valid records

OUTPUT
Validated Clean Data
(High Quality)

5 DATA CONSUMPTION FOR DASHBOARD

- Connect Looker Studio to Databricks
- Build dashboards
- KPIs & insights
- Filters & drill-downs

OUTPUT
Business Insights & Dashboards

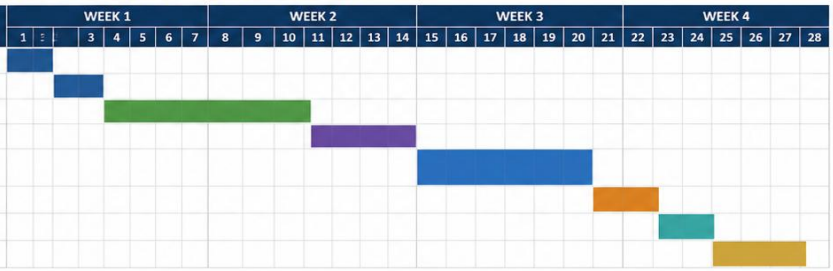
6 BUSINESS VALUE

- Data-driven decisions
- Monitor sales performance
- Identify high/low performers
- Optimize pricing vs MMR
- Improve data quality

OUTPUT
Better Decisions,
Higher Profitability

PROJECT TIMELINE – GANTT CHART

ID	PHASE / TASK	DURATION (DAYS)	START DATE	END DATE
1	Project Planning & Setup	2	01-May-25	02-May-25
2	Load CSV into Databricks	2	03-May-25	04-May-25
3	Data Transformation & Cleaning	7	05-May-25	11-May-25
4	Data Validation & Quality Checks	4	12-May-25	15-May-25
5	Connect to Looker Studio & Build Dashboard	6	16-May-25	21-May-25
6	Testing & Review	2	23-May-25	23-May-25
7	Deployment & Handover	2	24-May-25	25-May-25
8	Go-Live & Monitoring	3	26-May-25	28-May-25



PROJECT GOAL

Clean, validate and analyze car sales data to deliver actionable insights through dashboards.

KEY DELIVERABLES

- Cleaned Table: car_sales_clean
- Data Quality Report
- Looker Studio Dashboard

SUCCESS METRICS

- 100% Data Loaded
- Duplicate Records Flagged
- Data Quality > 95%
- Dashboard Live & Adopted

STAKEHOLDERS

- Data Analyst / Engineer
- Business Intelligence Team
- Management / Decision Makers

TOOLS & TECHNOLOGIES

databricks

Looker Studio